

Exp No: 26

TRANSACTION LEVEL MODELING (TLM) USING SYSTEMC

```
GNU nano 8.7.1 tlm.cpp
#define SC_INCLUDE_DYNAMIC_PROCESSES

#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>

using namespace sc_core;
using namespace tlm;
using namespace std;

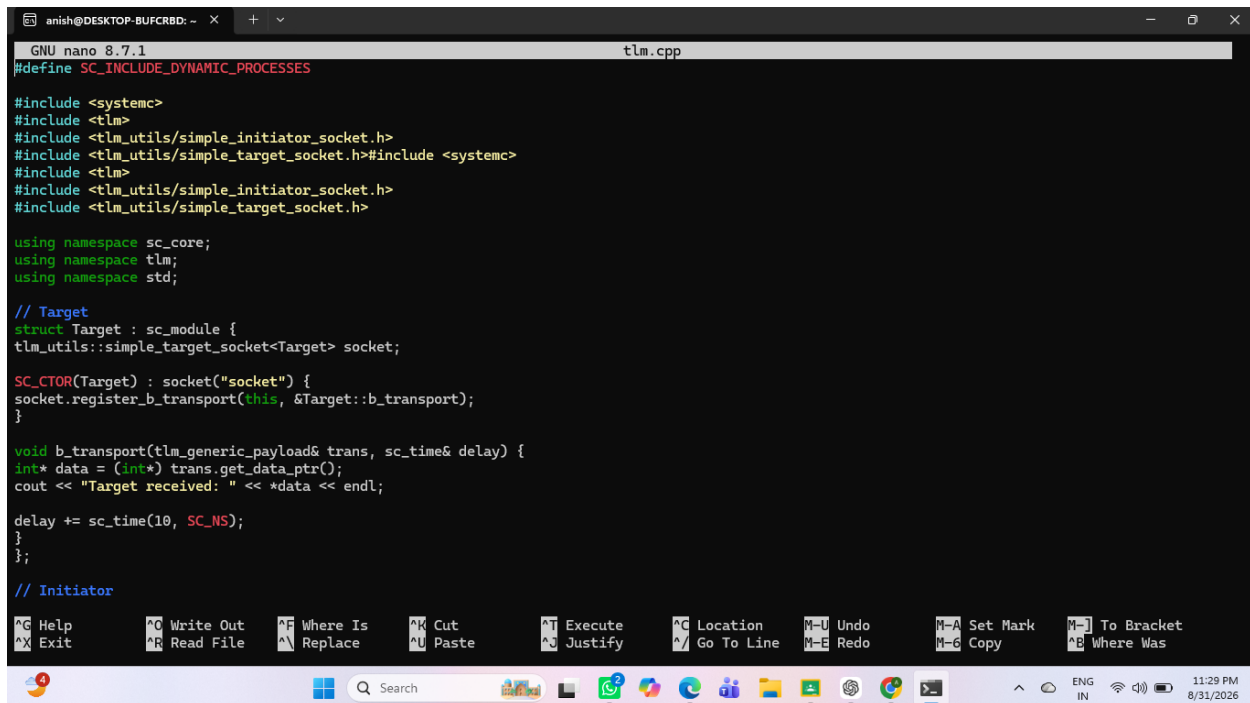
// Target
struct Target : sc_module {
    tlm_utils::simple_target_socket<Target> socket;

    SC_CTOR(Target) : socket("socket") {
        socket.register_b_transport(this, &Target::b_transport);
    }

    void b_transport(tlm_generic_payload& trans, sc_time& delay) {
        int* data = (int*) trans.get_data_ptr();
        cout << "Target received: " << *data << endl;

        delay += sc_time(10, SC_NS);
    }
};

// Initiator
```



```
GNU nano 8.7.1 CMakeLists.txt
cmake_minimum_required(VERSION 3.10)

project(TLM_Experiment)

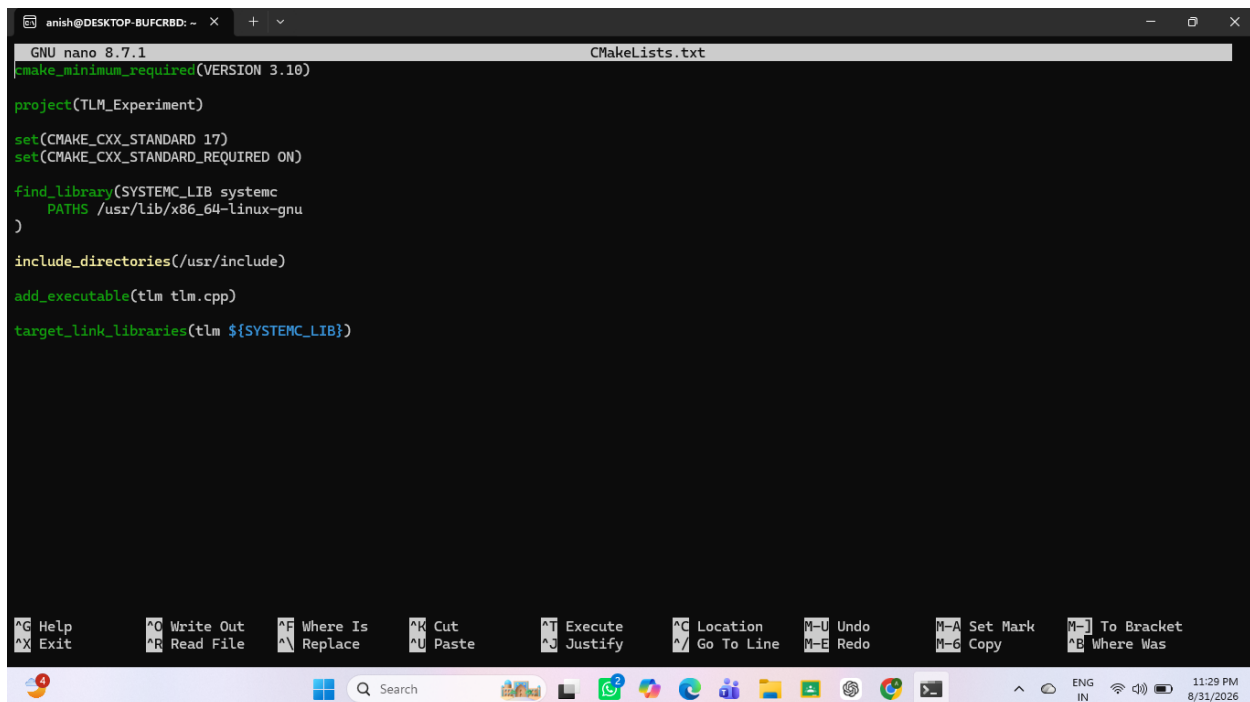
set(CMAKE_CXX_STANDARD 17)
set(CMAKE_CXX_STANDARD_REQUIRED ON)

find_library(SYSTEMC_LIB systemc
    PATHS /usr/lib/x86_64-linux-gnu
)

include_directories(/usr/include)

add_executable(tlm tlm.cpp)

target_link_libraries(tlm ${SYSTEMC_LIB})
```



```
anish@DESKTOP-BUFCRBD: ~  
anish@DESKTOP-BUFCRBD:~/systemc_test$ cd build  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ cmake .. -DCMAKE_CXX_COMPILER=g++-14  
-- The C compiler identification is GNU 15.2.0  
-- The CXX compiler identification is GNU 14.3.0  
-- Detecting C compiler ABI info  
-- Detecting C compiler ABI info - done  
-- Check for working C compiler: /usr/bin/cc - skipped  
-- Detecting C compile features  
-- Detecting C compile features - done  
-- Detecting CXX compiler ABI info  
-- Detecting CXX compiler ABI info - done  
-- Check for working CXX compiler: /usr/bin/g++-14 - skipped  
-- Detecting CXX compile features  
-- Detecting CXX compile features - done  
-- Configuring done (4.2s)  
-- Generating done (0.0s)  
-- Build files have been written to: /home/anish/systemc_test/build  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ make  
[ 50%] Building CXX object CMakeFiles/tlm.dir/tlm.cpp.o  
/home/anish/systemc_test/tlm.cpp:6:44: warning: extra tokens at end of #include directive  
6 | #include <tlm_utils/simple_target_socket.h>#include <systemc>  
|                                           ^  
[100%] Linking CXX executable tlm  
[100%] Built target tlm  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ ./tlm  
  
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46  
Copyright (c) 1996-2025 by all Contributors,  
ALL RIGHTS RESERVED  
Initiator sending: 555  
Target received: 555  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ |
```